This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

HySeg: Learning Generative Priors for Structure-Aware Remote Sensing Segmentation

Jie Qiu[1], Xin Li[2], Fan Yang[2], Yan Wang[1], Dong Yu[3], Changying Wang[1],
Linwei Dai[4], Yongxiang Chen[1], Youqin Chen[5,*], Jianzhang Chen[1,*]
[1]Fujian Agriculture and Forestry University [2]Alpaca AI Lab
[3]Beijing Jiaotong University [4]iFLYTEK [5]Fujian University of Technology

Abstract

High-resolution remote sensing imagery exhibits complex spatial regularities where topology, continuity, and region adjacency govern semantic organization. However, existing remote sensing image semantic segmentation (RSISS) networks, being predominantly discriminative, estimate strong posteriors from data while lacking generative priors that encode such structural dependencies. This imbalance leads to fragmented boundaries, texture overfitting, and poor cross-dataset generalization. We address this challenge by reformulating RSISS as posterior inference grounded in generative structural priors, introducing **HySeg***, a hybrid generative–discriminative segmentation paradigm that learns structure-consistent priors through generative modeling and guides posterior inference for remote sensing segmentation. At its core, the MeanStruct module, a MeanFlow-based generative prior learner, models semantic topology as a continuous stochastic field, while the Prior-to-Affinity Projection (P2A) dynamically transforms this field into topology-aware, class-conditional affinities that guide posterior inference in the Dynamic Affinity-driven Segmentation (DAS) head. Our approach is model-agnostic and seamlessly integrates with diverse backbones, consistently improving structural coherence and generalization. Across four challenging RSISS benchmarks, HySeg achieves state-of-the-art performance and advances remote sensing segmentation from appearance-based perception to structural reasoning. Our code is available at* https://github.com/HeryJie/HySeg.

1. Introduction

Remote sensing image semantic segmentation (RSISS) lies at the heart of large-scale Earth observation, converting satellite and aerial imagery into pixel-level maps that reveal land-cover patterns and human activity. As a foundation for urban infrastructure mapping, environmental surveillance, and disaster assessment [13, 37, 49], RSISS transforms raw pixels into structured spatial knowledge. Yet the task is uniquely challenging: unlike natural images, remote sensing scenes exhibit strong spatial heterogeneity, intricate topological interconnections, and context-dependent semantics shaped by geographic continuity and region adjacency.

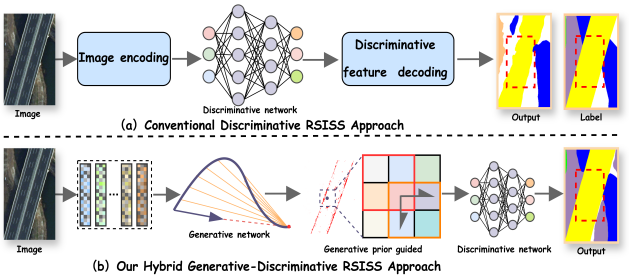


Figure 1. **Idea illustration.** (a) Conventional discriminative RSISS models estimate posteriors without modeling structural priors, resulting in fragmented boundaries and class confusion. (b) Our HySeg couples image-conditioned generative priors with discriminative inference, enforcing topology-aware coherence and globally consistent segmentation. Best viewed in colors.

Recent advances in RSISS have been driven by increasingly sophisticated discriminative architectures. CNN-based models such as DeepLab [4, 5, 58], HRNet [54], and PSPNet [63] capture multi-scale context through atrous convolution and pyramid pooling, forming the backbone of most high-resolution RS pipelines. Transformer-based architectures (*e.g.*, SegFormer [60], Swin-Unet [2], and scale-adaptive designs [29]) further enhance global context modeling and long-range spatial reasoning, achieving remarkable performance across diverse RS benchmarks. Beyond traditional closed-set settings, language-guided and open-vocabulary segmentation methods [3, 21, 26] extend semantic generalization across heterogeneous domains, while semi- and self-supervised strategies [41, 53, 67] improve robustness under sparse supervision. Related scene pars-

*Corresponding authors: chenyouqin@fjut.edu.cn, jchen@fafu.edu.cn

ing studies also explore mining supportive knowledge to enhance robustness and generalization [34]. However, despite architectural sophistication, existing approaches remain fundamentally appearance-driven, adept at delineating textures and boundaries but limited in capturing the relational structures that govern spatial organization. This imbalance between perception and reasoning exposes a fundamental gap in existing RSISS methodologies: while the progress has been made in *seeing* but not in *understanding*, the shift from recognition to structural reasoning remains unfinished.

To close this gap, we move beyond appearance-driven learning and recast RSISS as posterior inference constrained by structural priors (see Fig. 1). Building on this insight, we introduce **HySeg**, a hybrid generative–discriminative paradigm that learns structure-consistent priors via generative modeling and integrates them into posterior inference. This design explicitly couples generative priors with discriminative posteriors, turning remote sensing segmentation into a structured reasoning process that unifies local appearance cues with global spatial organization, thereby yielding predictions that are spatially coherent and topologically consistent. Yet a central question remains: how can structural priors be learned so that they reflect the continuous geography and inter-region relations that shape remote sensing scenes?

We pursue this question from a generative perspective and introduce MeanStruct, a MeanFlow-based [12, 14] prior learner that models structure as a continuous transport process in label space. It estimates a mean-transport (average-velocity) field that maps simple structural prototypes to the empirical distribution of segmentation maps under supervision. By modelling *how* semantic topology evolves, rather than merely *which* labels appear, MeanStruct learns a probabilistic structural prior that can better capture the intrinsic geometry of real-world layouts. While MeanStruct provides a rich probabilistic description of structural topology, the learned prior remains abstract and must be translated into a form that can guide pixel-level reasoning. To this end, we propose the Prior-to-Affinity Projection (P2A), which operationalizes the generative prior by translating it into topology-aware, class-conditional affinity kernels that regulate pixel interactions and enforce structural coherence across regions. Acting as a structural translator, P2A bridges generative reasoning and discriminative inference, transforming the implicit topology encoded by the prior into an explicit spatial mechanism through which structural knowledge governs semantic organization.

To integrate these cues into posterior inference, we introduce a Dynamic Affinity-driven Segmentation (DAS) head that leverages the projected affinities to guide message passing among pixels, aligning appearance with structural topology. Built as a generative–discriminative hybrid, our **HySeg** is backbone-agnostic and can be combined with diverse backbones, consistently improving structural coherence and generalization. Our main **contributions** are threefold:

- **Paradigm Shift.** We reformulate remote sensing image semantic segmentation (RSISS) as posterior inference constrained by generative structural priors, bridging perception and reasoning through a unified generative–discriminative paradigm.
- **Hybrid Approach.** We propose **HySeg**, a novel hybrid generative–discriminative framework that unifies generative prior learning (MeanStruct) and discriminative inference through Prior-to-Affinity Projection (P2A) and Dynamic Affinity-driven Segmentation (DAS), enabling structure-aware posterior reasoning in segmentation.
- **SOTA Performance.** Evaluated across four challenging RSISS benchmarks, our **HySeg** achieves state-of-the-art (SOTA) performance, consistently improving structural coherence, spatial consistency, and cross-dataset generalization over existing methods.

2. Related Work

Remote Sensing Image Semantic Segmentation. RSISS is a core task in geospatial vision that converts aerial or satellite imagery into dense land-cover maps [43]. CNN-based models such as DeepLab [4, 5, 58], HRNet [54], PSPNet [63], EFRNet [23], and EKENet [33] leverage atrous convolution, feature reconstruction, and knowledge enhancement to capture multi-scale context, forming the foundation of modern high-resolution semantic segmentation pipelines. Transformer-based designs (*e.g.*, SegFormer [60], Swin-Unet [2], RSSFormer [61], scale-adaptive models [29], and RS-vHeat [16]) further improve global context modeling and long-range reasoning, achieving state-of-the-art results on diverse benchmarks. Beyond closed-set settings, recent vision–language models such as RMSIN [26], SegEarth-OV [21], and related formulations [18, 48, 62] enable category-agnostic understanding. Complementary research in domain adaptation and generalization [7, 27, 52] mitigates cross-sensor and regional shifts, while self- and weakly supervised methods [1, 19, 66] reduce annotation demands by exploiting structural cues in unlabeled data. Generative paradigms, including diffusion-based methods [42, 65, 68], further advance RSISS by learning data distributions and representation alignment. In contrast, our approach unifies generative and discriminative paradigms within a single inference framework that learns structural priors through generation and refines them via discrimination.

MeanFlow and Generative Transport Modeling. Generative transport modeling views data generation as learning continuous velocity fields that move samples from a prior to the data manifold, offering a physically interpretable alternative to stochastic diffusion. Flow Match-

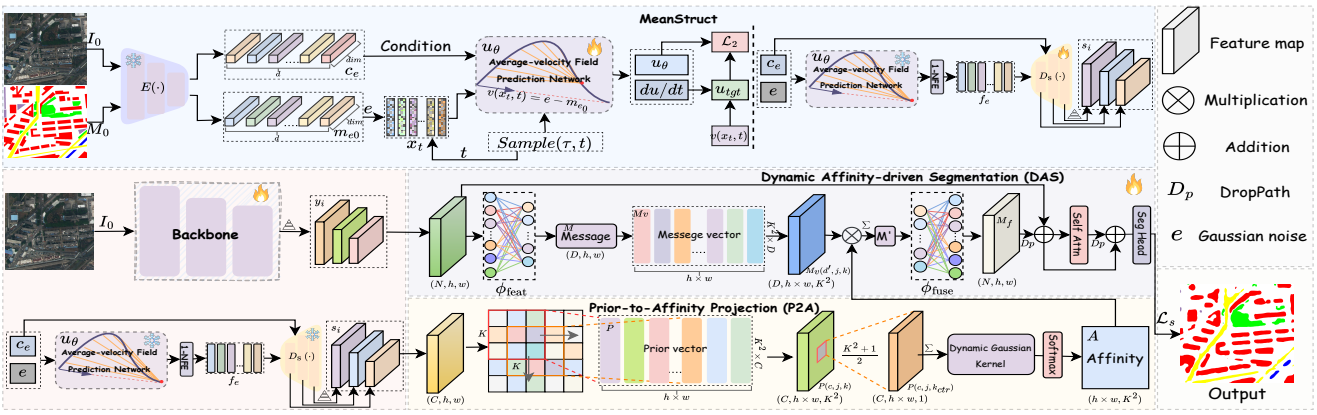


Figure 2. **Overview of HySeg.** Our framework formulates RSISS as posterior inference constrained by generative structural priors. Stage I (**MeanStruct**) learns structure-consistent priors via MeanFlow; Stage II (**DAS+P2A**) leverages these priors for topology-aware message passing and segmentation. See Sec. 3 for details.

ing [10, 20, 25] directly regresses time-dependent velocity fields, while Rectified Flow [28] enforces nearly linear transport for stable and efficient sampling. These formulations replace stochastic noise with structured, geometry-preserving trajectories. Building on this foundation, MeanFlow [12] extends flow matching to model the expected (mean-field) dynamics, describing how distributions evolve on average rather than through individual stochastic paths. The resulting smooth and efficient transport makes MeanFlow well suited for learning structural priors [64] that encode spatial topology in a geometry-aware manner. Similar generative formulations have also shown effectiveness in dense visual prediction tasks such as optical flow estimation [35, 36]. Building on this property, our method models structural evolution generatively and fuses it into discriminative inference for topology-aware segmentation.

3. Method

Existing RSISS models are mainly discriminative, focusing on appearance cues while ignoring structural dependencies. We instead formulate segmentation as posterior inference constrained by generative structural priors. Our approach couples a generative prior learner with a discriminative segmentor guided by learned dynamic affinities (Fig. 2). Specifically, Stage I (**MeanStruct**) models semantic topology as a continuous flow field, learning structure-consistent priors through a one-step MeanFlow update. Stage II translates the learned priors into topology-aware affinities through **P2A**, which regulate prior-guided message passing in **DAS**. The overall design is backbone-agnostic and generalizes well across architectures.

3.1. MeanStruct Module

Given an RGB image $I_0 \in \mathbb{R}^{H\times W\times 3}$ and its ground-truth mask $M_0 \in \{0,\dots,C_0{-}1\}^{H\times W}$ (used *only* during training), we learn a dense structural prior $f_e \in \mathbb{R}^{h\times w\times d}$, where s denotes the spatial downsampling rate, $h{=}H/s$ and $w{=}W/s$ denote the downsampled resolution, and d denotes the embedding dimension. The prior *explicitly* encodes topological continuity and regional adjacency, serving as a generative constraint that regularizes posterior inference. We formulate this as an image-conditioned generation task and employ a MeanFlow-based single-step (*1-NFE*) learner, where a conditional vector field transports a perturbed embedding to a structure-consistent prior in one update, balancing structural fidelity and efficiency.

Encoders and Path Construction. We extract two embeddings: an image-conditioned context $c_e = E(I_0) \in \mathbb{R}^{h\times w\times d}$ and a target prior embedding $m_{e_0} = E(M_0) \in \mathbb{R}^{h\times w\times d}$, where M_0 (used *only* during training) provides the ground-truth mask, and c_e serves as a *guidance signal* conditioning the transport field. We then define a linear path in the prior space:

$$x_t = (1-t)\, m_{e_0} + t\, e, \tag{1}$$

where $t \in [0,1]$ and $e \sim \mathcal{N}(0, \mathbf{I})$ denotes standard Gaussian noise.

MeanFlow-based Averaged Velocity Learning. We follow MeanFlow [12] to learn a continuous transport field that links noisy embeddings to structure-consistent priors. A learnable vector field $u_\theta(x_t, \tau, t)$ models the *interval-averaged velocity* between two latent states x_τ and x_t:

$$u(x_t,\tau,t) = \frac{1}{t-\tau}\int_\tau^t v(x_\eta,\eta)\, d\eta. \tag{2}$$

For the linear interpolation path $x_t = (1-t)m_{e_0} + t\, e$, the instantaneous velocity is constant, $v(x_t,t) = \frac{d}{dt}x_t = e - m_{e_0}$.

(1) MeanFlow identity. The MeanFlow identity relates the averaged and instantaneous velocities:

$$u(x_t,\tau,t) = v(x_t,t) - (t-\tau)\frac{d}{dt}u(x_t,\tau,t), \tag{3}$$

where

$$\frac{d}{dt}u(x_t,\tau,t) = \partial_x u(x_t,\tau,t)\cdot v(x_t,t) + \partial_t u(x_t,\tau,t).$$

(2) Training objective. Using Eq. (3), the target velocity is

$$u_{\text{tgt}} = v(x_t,t) - (t-\tau)\Big(\partial_x u_\theta(x_t,\tau,t)\cdot v(x_t,t) + \partial_t u_\theta(x_t,\tau,t)\Big), \tag{4}$$

where $\partial_x u_\theta \cdot v$ and $\partial_t u_\theta$ are computed via automatic differentiation (JVP) [12]. The training loss is:

$$\mathcal{L}_{\text{MeanStruct}} = \mathbb{E}_{t,\tau}\left[\|u_\theta(x_t,\tau,t) - \text{sg}(u_{\text{tgt}})\|_2^2\right], \tag{5}$$

where $\text{sg}(\cdot)$ denotes stop-gradient. This formulation encourages u_θ to approximate the mean transport dynamics that preserve structural topology. Geometrically, the learned MeanFlow field induces a smooth, low-divergence vector transport on the semantic manifold, which tends to preserve adjacency relationships and local connectivity under mild smoothness and Lipschitz assumptions on the velocity field [12, 25]. Experimentally, we observe that such continuous flows produce priors f_e that encode spatial continuity and region adjacency more reliably than purely discriminative features, providing a generative structural bias for subsequent posterior inference rather than a strict topological guarantee.

Efficient 1-NFE Prior Generation. During inference, the MeanFlow field performs a deterministic rollback that transports a random latent toward the structural manifold in a single function evaluation. From Eq. (2), the rollback trajectory is expressed as

$$x_\tau = x_t - (t-\tau)\,u(x_t,\tau,t), \tag{6}$$

and setting $(\tau,t) = (0,1)$ with $e \sim \mathcal{N}(0,\mathbf{I})$ yields

$$f_e = e - u_\theta(e,0,1). \tag{7}$$

This single-step transformation (1-NFE) produces a structurally coherent prior that encourages spatial continuity and regional adjacency while remaining computationally minimal. The resulting prior acts as a generative structural constraint that nudges posterior inference toward more geometrically consistent solutions, without imposing an explicit topological guarantee.

Spatial Decoder and Feature Alignment. To couple the generative prior with the discriminative backbone, a lightweight spatial decoder D_s hierarchically projects the global prior f_e into a multi-scale structural pyramid $S = \{s_i\}_{i=1}^3$:

$$s_i = D_s^{(i)}(f_e) \in \mathbb{R}^{c_i \times h_i \times w_i}. \tag{8}$$

Each prior scale s_i is aligned with the corresponding backbone feature y_i in spatial resolution—and, when necessary, in channel dimension through a 1×1 projection. These aligned priors act as conditional structural fields, embedding the learned topology into the feature hierarchy.

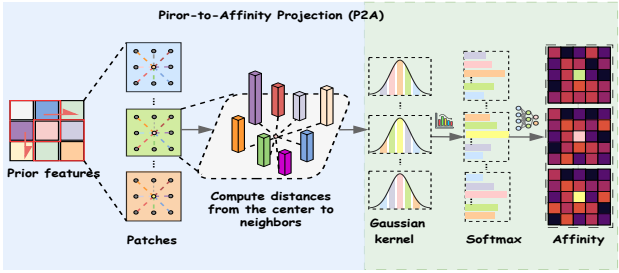


Figure 3. **Prior-to-Affinity Projection.** P2A converts learned priors into topology-aware affinities for DAS. It unfolds local $K\times K$ neighborhoods, measures channel-wise center–neighbor discrepancies, converts them into similarity weights via a learnable Gaussian kernel, and normalizes them using masked softmax.

3.2. Prior-to-Affinity Projection (P2A)

MeanStruct generates image-conditioned priors that capture spatial continuity and regional adjacency but remain implicit. To make them usable, **P2A** (Fig. 3) converts these learned prior features into pixel-wise, topology-aware affinities by projecting local $K \times K$ neighborhoods into an affinity field for message passing. This transformation turns generative priors into explicit structural weights that constrain the discriminative head, improving boundary accuracy and intra-region consistency while remaining lightweight and backbone-agnostic.

For a prior feature map $s_i \in \mathbb{R}^{C\times h\times w}$ at scale i, let $P \in \mathbb{R}^{C\times L\times K^2}$ denote the unfolded local neighborhood tensor, where $L = h \times w$ and K is the neighborhood size (see details in the *Supplementary Material*). For each site j, let $p_j^{(c)} = P(c,j,k_{\text{ctr}})$ denote the center feature at channel c, and let $x_j^{(c,k)} = P(c,j,k)$ denote its k-th neighboring feature. Their discrepancy is defined as

$$D_j^{(c,k)} = \left(p_j^{(c)} - x_j^{(c,k)}\right)^2. \tag{9}$$

We then aggregate channel-wise discrepancies and convert them into normalized affinity weights:

$$A_j^{(k)} = \text{Softmax}_k\left(-\frac{\sum_{c=1}^{C} D_j^{(c,k)}}{2\sigma^2 + \xi}\right), \tag{10}$$

where σ is a learnable bandwidth and ξ is a small stability constant. This yields the affinity field

$$A = \{A_j^{(k)}\} \in \mathbb{R}^{L\times K^2}, \qquad \sum_{k=1}^{K^2} A_j^{(k)} = 1. \tag{11}$$

The resulting affinity field provides structurally informed weights for local message passing in the DAS head. Detailed definitions of neighborhood unfolding, spatial index mapping, boundary handling, and masked normalization are given in the *Supplementary Material*. Unlike self-attention, P2A derives affinities from the learned structural

prior rather than from pairwise similarities among segmentation features.

3.3. Dynamic Affinity-driven Segmentation (DAS)

Within the posterior inference stage, the Dynamic Affinity-driven Segmentation (DAS) head performs discriminative reasoning under the guidance of generative priors. The topology-aware affinities A derived from P2A serve as structural constraints that modulate feature propagation, ensuring that local evidence aggregation respects the continuity and adjacency encoded in the prior. In this formulation, DAS realizes posterior inference regularized by the generative prior field f_e, aligning discriminative decoding with structural consistency.

Message Projection and Aggregation. Given backbone features $y_i \in \mathbb{R}^{N\times h\times w}$ aligned with the prior scale s_i, a 1×1 projection $\phi_{\text{feat}} : \mathbb{R}^N \to \mathbb{R}^D$ produces latent representations $M = \phi_{\text{feat}}(y_i)$. Unfolding M into $K\times K$ neighborhoods yields local evidence $M_v \in \mathbb{R}^{D\times L\times K^2}$, which is aggregated via affinity-guided weighting:

$$M'(d', j) = \sum_{k=1}^{K^2} A_j^{(k)} M_v(d', j, k), \tag{12}$$

where $A_j^{(k)}$ encodes the structural prior probability of information exchange between pixels. This process approximates a single update step of the posterior field, where each pixel refines its latent representation by marginalizing over structurally correlated neighbors. The result is fused through a lightweight MLP $\phi_{\text{fuse}} : \mathbb{R}^D \to \mathbb{R}^N$, producing the prior-regularized latent map $M_f = \phi_{\text{fuse}}(M') \in \mathbb{R}^{N\times h\times w}$.

Global Contextualization and Decoding. To extend posterior reasoning beyond local neighborhoods, DAS adopts a dual-residual design that integrates prior-guided aggregation and self-attention refinement:

$$y_i' = y_i + D_p(M_f), \qquad y_{\text{out}} = y_i' + D_p\big(\text{SA}(y_i')\big), \tag{13}$$

where SA denotes multi-head self-attention over spatial tokens and D_p (DropPath) regularizes the residual updates. Finally, y_{out} is decoded into per-pixel logits, yielding a posterior estimate constrained by the learned structural priors. The segmentation branch is optimized using a standard loss $\mathcal{L}_{\text{s}} = \mathcal{L}_{\text{RSISS}}$ [24], completing the generative–discriminative inference loop.

4. Experiments

4.1. Experimental Setup

Datasets and Evaluation Metrics. We evaluate on four benchmarks: LoveDA [55], UAVid [38], ISPRS Potsdam, and ISPRS Vaihingen [44]. LoveDA, with a ground sampling distance (GSD) of 30 cm, covers seven land-cover classes from urban and rural scenes. UAVid provides 420 high-resolution UAV images (3840×2160 or 4096×2160) across eight urban categories, following the splits in [57]. Vaihingen (9 cm GSD) and Potsdam (5 cm GSD) each contain six classes, following the splits in Wang et al. [57]. We adopt standard metrics including overall accuracy (OA), mean intersection-over-union (mIoU), and mean F1-score (mF1) to measure pixel accuracy, class-wise balance, and region overlap [8, 9, 32].

Implementation Details. To validate the generality of HySeg, we integrate it into multiple state-of-the-art segmentation frameworks, including UNetFormer [57] (ResNet18/34/50 [15]), LSKNet [24] (LSKNet_T/S), DCSwin [56] (Swin_T/B/S [30]), and D2LS [69] (ConvNeXt_B [31]). This setup ensures coverage of both CNN- and Transformer-based paradigms, from lightweight to large-scale backbones. All experiments are implemented in PyTorch and trained on a single NVIDIA A800 GPU using AdamW. We follow the original training settings of each backbone [24, 57] for fair comparison across datasets (LoveDA, UAVid, Potsdam, Vaihingen). During inference, multi-scale testing and horizontal flipping are applied to enhance robustness.

4.2. Main Results

Quantitative Evaluation. Results summarized in Tab. 1 show consistent improvements on LoveDA, UAVid, and ISPRS Potsdam/Vaihingen across multiple backbones, including UNetFormer (ResNet18/34/50), DCSwin (Swin_T/S/B), LSKNet (LSKNet_T/S), and D2LS (ConvNeXt_B). Incorporating HySeg consistently boosts mIoU relative to purely discriminative counterparts. On LoveDA, nearly all categories benefit, with notable gains of +5.0 mIoU on *Barren* (Swin_B) and +3.2 on *Agriculture* (LSKNet_S). Similar improvements are observed on ISPRS Vaihingen, Potsdam, and UAVid, where HySeg increases mIoU by +2.2, +1.7, and +1.6, respectively. The largest gains occur in structure-dominated categories (*Building*, *Barren*, *Forest*), reflecting improved boundary precision and intra-region coherence. These findings validate that converting generative structural priors (MeanStruct) into topology- and class-aware affinities via P2A, and propagating them through DAS, effectively injects structural reasoning into discriminative segmentation, yielding robust, cross-architecture, and cross-dataset gains. Beyond region-overlap metrics, we further evaluate structural fidelity using topology-related measures, including clDice [46] and Betti-based topology errors [17, 51]. As shown in Tab. 2, HySeg consistently improves clDice and reduces both β_0- and β_1-Errors across representative backbone-matched settings, indicating better preservation of connectivity and region-level topology. Such topology-related metrics have also been used in prior topology-preserving segmentation stud-

Table 1. **Comparison with SOTA methods** on the LoveDA, Vaihingen, Potsdam and UAVid datasets (partial). △ denotes the performance improvement achieved by enhancing the base model with HySeg. The best results are highlighted in **bold**.

Method	LoveDA								Vaihingen			Potsdam			UAVid
	Backbone	Bkg	Bld	Rd	Wat	Bar	For	Agr	mIoU↑	mF1↑	OA↑	mIoU↑	mF1↑	OA↑	mIoU↑	mIoU↑
Segmenter[50]	ViT-B	38.0	50.7	48.7	77.4	13.3	43.5	58.2	47.1	81.9	85.3	76.1	82.5	82.2	80.6	66.0
MANet[22]	ResNet50	38.7	51.7	42.6	72.0	15.3	42.1	57.7	45.7	90.8	91.3	83.3	92.5	91.1	86.3	62.6
DeepLabV3+[6]	ResNet50	43.0	50.9	52.0	74.4	10.4	44.2	58.5	47.6	81.2	86.2	79.6	87.9	86.2	78.7	60.6
CTMFNet[47]	ResNet50	46.8	59.4	54.8	74.4	22.6	45.2	60.8	52.0	91.1	91.6	83.9	93.0	91.5	87.1	67.4
DANet[11]	ResNet50	46.4	56.8	57.0	79.3	14.6	46.6	64.6	52.2	85.9	89.2	76.0	90.4	89.8	82.7	60.6
SAM_RS[39]	ResNet18	46.3	60.4	54.5	77.8	20.2	47.1	61.2	52.5	90.1	91.2	82.9	92.2	90.4	85.9	63.8
LOGCAN++[40]	ResNet50	47.4	58.4	56.5	80.1	18.4	48.0	64.8	53.4	91.4	91.8	84.3	92.9	91.5	87.0	68.8
In2NeCT[45]	ResNet18	46.1	60.8	57.5	80.5	24.0	47.7	63.8	54.3	91.3	91.5	84.5	93.2	91.8	87.9	/
UNetFormer[57]	ResNet18	44.7	58.8	54.9	79.6	20.1	46.0	62.5	52.4	90.0	90.9	82.0	92.2	90.7	85.7	67.8
	ResNet34	44.8	57.8	54.9	79.1	18.5	46.4	62.4	52.0	90.3	91.1	82.5	92.4	91.0	86.1	68.0
	ResNet50	45.8	58.5	57.1	79.3	17.0	46.3	63.7	52.5	90.6	91.3	83.1	92.6	91.0	86.4	68.5
Ours	ResNet18	47.5	61.1	56.5	81.5	22.3	47.3	65.0	54.5	90.7	91.5	82.8	92.7	91.3	86.5	68.7
	△	+2.8	+2.3	+1.6	+1.9	+2.2	+1.3	+2.5	+2.1	+0.7	+0.6	+0.8	+0.5	+0.6	+0.8	+0.9
	ResNet34	46.8	59.1	57.2	81.1	19.3	47.4	64.1	53.6	91.0	91.8	83.2	92.8	91.3	86.7	68.8
	△	+2.0	+1.3	+2.3	+2.0	+0.8	+1.0	+1.7	+1.6	+0.7	+0.7	+0.7	+0.4	+0.3	+0.6	+0.8
	ResNet50	**48.6**	60.8	59.3	82.0	20.1	46.5	64.9	54.7	91.6	92.1	84.4	93.0	91.5	86.9	69.5
	△	+2.8	+2.3	+2.2	+2.7	+3.1	+0.2	+1.2	+2.2	+1.0	+0.8	+1.3	+0.4	+0.5	+0.5	+1.0
DCSwin[56]	Swin_T	44.8	59.9	57.9	81.0	17.5	47.8	61.6	52.9	90.0	91.2	82.1	92.0	90.1	85.3	68.0
	Swin_S	45.1	60.8	57.7	82.1	18.2	48.5	61.5	53.4	90.6	91.6	83.1	92.7	91.2	87.1	68.4
	Swin_B	44.7	61.6	57.4	81.4	17.1	48.2	59.9	52.9	90.5	91.7	83.4	92.9	91.3	86.9	68.1
Ours	Swin_T	46.8	60.0	57.0	81.6	19.8	48.0	64.3	53.9	90.5	92.2	82.9	93.3	91.9	87.7	68.9
	△	+2.0	+0.1	-0.9	+0.6	+2.3	+0.2	+2.7	+1.0	+0.5	+1.0	+0.8	+1.3	+1.8	+2.4	+0.9
	Swin_S	45.8	61.5	57.3	**82.7**	20.6	48.8	61.3	54.0	91.3	93.1	84.3	93.6	92.2	88.1	69.4
	△	+0.7	+0.7	-0.4	+0.6	+2.4	+0.3	-0.2	+0.6	+0.7	+1.5	+1.2	+0.9	+1.0	+1.0	+1.0
	Swin_B	46.3	**61.9**	57.7	81.5	22.1	48.7	63.0	54.5	92.0	93.5	85.4	93.6	92.2	88.1	69.7
	△	+1.6	+0.3	+0.3	+0.1	+5.0	+0.5	+3.1	+1.6	+1.5	+1.8	+2.0	+0.7	+0.9	+1.2	+1.6
LSKNet[24]	LSKNet_T	46.4	59.5	57.1	79.9	21.8	46.6	61.4	53.2	90.9	91.5	83.6	92.3	91.1	86.1	69.3
	LSKNet_S	46.7	59.9	58.3	80.2	**24.6**	46.4	61.8	54.0	91.0	91.7	83.7	93.1	92.0	87.2	70.0
Ours	LSKNet_T	47.0	60.2	59.4	81.5	23.9	48.4	63.2	54.8	91.5	92.9	84.6	93.4	91.9	87.2	69.8
	△	+0.6	+0.7	+2.3	+1.6	+2.1	+1.8	+1.8	+1.6	+0.6	+1.4	+1.0	+1.1	+0.8	+1.1	+0.5
	LSKNet_S	47.9	61.3	59.6	81.6	23.3	49.4	**65.0**	55.4	**92.2**	**93.8**	**85.9**	94.3	93.4	88.9	70.9
	△	+1.2	+1.4	+1.3	+1.4	-1.3	+3.0	+3.2	+1.4	+1.2	+2.1	+2.2	+1.2	+1.4	+1.7	+0.9
D2LS[69]	ConvNeXt_B	47.6	61.2	59.1	81.6	23.8	48.8	64.8	55.3	91.9	/	/	94.7	/	/	70.9
Ours	ConvNeXt_B	47.8	61.7	**59.7**	81.9	23.8	**49.7**	64.9	**55.6**	92.1	93.6	85.7	**94.8**	**93.9**	**89.5**	**71.4**
	△	+0.2	+0.5	+0.6	+0.3	+0.0	+0.9	+0.1	+0.3	+0.2	/	/	+0.1	/	/	+0.5

ies [59]. Additional results on modern pretrained RS backbones and comparisons with lightweight structural constraints are provided in the *Supplementary Material*.

Table 2. **Topology-related evaluation** on Potsdam. Beyond region-overlap metrics, HySeg consistently improves clDice and reduces Betti errors across representative backbone-matched settings, indicating better structural fidelity.

Method	Backbone	clDice↑	β_0-Error↓	β_1-Error↓
UNetFormer	ResNet_18	0.9168±0.0060	1.6448	4.6806
HySeg (Ours)	ResNet_18	0.9236±0.0179	1.2369	3.8405
DCSwin	Swin_S	0.9120±0.0041	1.5774	4.8095
HySeg (Ours)	Swin_S	0.9291±0.0142	1.0972	3.5726
LSKNet	LSKNet_S	0.9174±0.0047	1.2718	3.8988
HySeg (Ours)	LSKNet_S	0.9305±0.0106	1.0253	3.0548
D2LS	ConvNeXt_B	0.9110±0.0142	2.5218	4.5119
HySeg (Ours)	ConvNeXt_B	0.9350±0.0467	1.0033	2.4643

Qualitative Evaluation. Fig. 4 presents visual comparisons across LoveDA, UAVid, Potsdam, and Vaihingen, using representative backbones (UNetFormer_R18, DCSwin_S, LSKNet_S, and D2LS_ConvNeXt_B) with and without HySeg. Across all settings, HySeg consistently produces sharper boundaries, more distinct Forest–Agriculture separation, and fewer Tree–Building or Car–Impervious Surfaces confusions. On LoveDA, baseline models often generate roof overflows on factory complexes and blurred Forest–Agriculture transitions, whereas HySeg yields compact, closed factory footprints and well-delineated vegetation zones by projecting generative priors (MeanStruct) into topology-aware affinities via P2A and propagating them in DAS. On UAVid, HySeg preserves large-scale building continuity that baselines tend to fragment. On Pots-

Table 3. **Ablation results** of HySeg on UNetFormer with ResNet18/50, Swin_B, LSKNet_S, and ConvNeXt_B backbones on the LoveDA dataset.

Method/Backbone	ResNet18	ResNet50	Swin_B	LSKNet_S	ConvNeXt_B
Baseline (Backbone+Standard Head)	52.42	52.52	52.03	54.01	52.77
+ DAS(w/o Prior)	52.76	53.00	52.87	54.12	53.26
+ Simple Prior Fusion	53.42	53.28	53.41	54.17	54.04
+ MeanStruct+DAS (w/o P2A)	52.01	51.97	52.20	52.00	52.41
+ MeanStruct+P2A+DAS (w/o Affinity & Message)	53.53	53.67	53.67	54.71	54.57
+ MeanStruct +P2A+DAS (w/o Self Attention)	54.02	54.31	54.28	55.13	55.20
HySeg (Full Model)	**54.48**	**54.74**	**54.46**	**55.37**	**55.63**

Table 4. **Effect of generative priors vs. discriminative priors** on MeanStruct performance with ResNet18/50, Swin_B, LSKNet_S, and ConvNeXt_B backbones.

Method/Backbone	ResNet18	ResNet50	Swin_B	LSKNet_S	ConvNeXt_B
Baseline (Backbone+ Standard Head)	52.42	52.52	52.03	54.01	52.77
+ ResNet18 Prior	52.74	52.41	51.88	53.98	50.20
+ Swin_B Prior	53.18	52.88	53.02	53.29	53.31
+ LSKNet_S Prior	53.87	54.21	52.71	54.55	53.14
+ ConvNeXt_B Prior	53.16	53.94	53.35	54.80	54.99
+ MeanStruct Prior	**54.48**	**54.74**	**54.46**	**55.37**	**55.63**

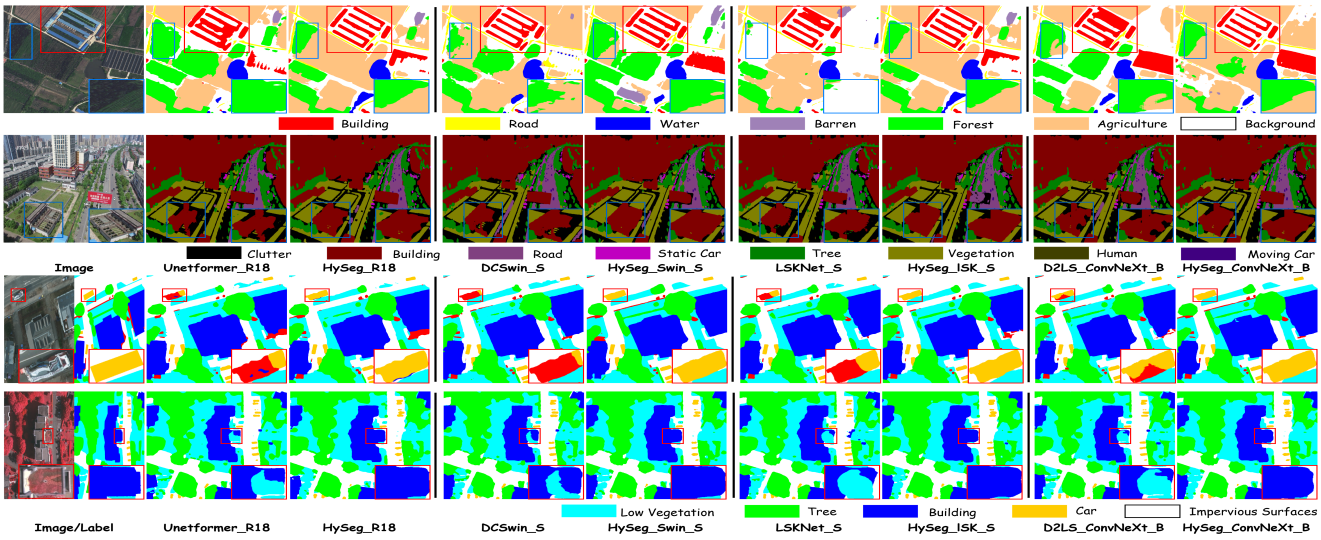


Figure 4. **Visual comparison** on LoveDA, UAVid, Potsdam, and Vaihingen using four backbones—UNetFormer_R18, DCSwin_S, LSKNet_S, and D2LS_ConvNeXt_B. Highlighted regions show that HySeg produces sharper boundaries, clearer Forest–Agriculture separation, and fewer Tree–Building and Car–Impervious Surfaces confusions.

Table 5. **Analysis of P2A affinity computation** on HySeg performance with ResNet18/50, Swin_B, LSKNet_S, and ConvNeXt_B backbones.

Method/Backbone	ResNet18	ResNet50	Swin_B	LSKNet_S	ConvNeXt_B
σ=0.5	53.82	53.90	53.68	54.46	54.20
σ=1.0	53.28	53.92	54.21	54.47	54.89
σ=1.5	53.40	53.50	53.35	54.45	55.23
σ=2.0	52.92	53.97	53.89	54.02	54.87
σ=learnable	**54.48**	**54.74**	**54.46**	**55.37**	**55.63**
$K = 3$	53.94	53.21	53.50	54.11	54.12
$K = 5$	54.44	54.70	54.42	55.19	55.24
$K = 7$	**54.48**	54.74	**54.46**	**55.37**	**55.63**
$K = 9$	54.35	**54.77**	54.30	55.26	55.40
num_heads=2	53.92	54.07	54.21	54.49	54.99
num_heads=4	54.03	54.43	54.27	54.36	55.56
num_heads=8	**54.48**	54.74	**54.46**	**55.37**	**55.63**
num_heads=16	54.42	**54.75**	54.39	55.35	55.50

dam and Vaihingen, it restores clear car contours and roof boundaries, effectively reducing Building–Tree and Car–Impervious Surfaces misclassifications. Overall, HySeg enhances structural integrity, enforces boundary coherence, and mitigates semantic ambiguity across diverse urban and rural scenes. Additional visual analyses are provided in the *Supplementary Material*.

4.3. Ablation Study

We conduct ablation experiments on the LoveDA dataset to quantify the contribution of each HySeg component. Four representative backbones are evaluated—UNetFormer (ResNet18/50), LSKNet (LSKNet_S), DCSwin (Swin_B), and D2LS(ConvNeXt_B). Results are reported as mIoU (↑) on the LoveDA *online testing* server under identical training settings as the baselines, ensuring that the observed gains originate solely from HySeg's architecture.

Component-Wise Analysis. As summarized in Tab. 3, the ablation on LoveDA reveals how HySeg components complement each other across five backbones (ResNet-18/50, Swin_B, LSKNet_S, ConvNeXt_B). The baseline performance gradually improves as additional modules are introduced. While DAS alone brings only modest gains, fusing MeanStruct priors already enhances mIoU, showing that generative cues are beneficial even without explicit affinity modeling. Once P2A is enabled, priors are properly translated into class- and topology-aware affinities, leading to a substantial leap. Disabling affinity propagation (*MeanStruct + P2A + DAS w/o Affinity & Message*) causes a pronounced decline, highlighting the importance of message passing for structural reasoning. Removing self-attention leads to only mild degradation, implying that local propagation guided by priors dominates the learning process. The complete HySeg consistently yields the strongest results, underscoring the effectiveness of its unified generative–discriminative design.

Ablation Studies on Key Designs and Hyperparameters. Tab. 4 shows ablation results on the LoveDA dataset using five backbones, including ResNet18, ResNet50, Swin_B, LSKNet_S, and ConvNeXt_B, measured by mIoU. The baseline (Backbone + Standard Head) serves as the reference. Adding different priors consistently boosts performance; for example, the ResNet18 prior raises its mIoU from 52.42 to 52.74, and similar gains are observed for Swin_B. The MeanStruct prior yields the largest improvement across all backbones, reaching 54.48 for ResNet18 and 55.63 for ConvNeXt_B. These results highlight the importance of generative priors, particularly MeanStruct, in enhancing structural coherence and segmentation accuracy.

As shown in Tab. 5, we conduct ablations on key hyperparameters, *i.e.,* σ, K, and the number of attention heads. For the Gaussian bandwidth σ, dynamic learning yields the

Table 6. **Efficiency–accuracy trade-off.** Evaluated with batch size 1 and 512×512 inputs on an NVIDIA A800 GPU. Both accuracy and computational cost are reported.

Method/Backbone	FPS ↑	GFlops ↓	Params(M) ↓	mIoU ↑
(1) Pure Discriminative Family				
UNetFormer_R18	188.12	23.48	11.72	52.42
UNetFormer_R50	124.85	48.36	24.24	52.52
LSKNet_S	84.69	31.46	14.39	54.01
D2LS_ConvNeXt_B	47.47	186.51	94.03	55.27
(2) Pure Generative Family				
Conditional Diffusion Model (10-NFE)	5.03	527.14	320.17	/
Conditional Flow-based Model (10-NFE)	10.24	260.33	108.22	/
UNet2DModel + DPM-Solver (20-NFE)	3.68	714.10	38.83	/
UNet2DModel + DPM-Solver (1-NFE)	40.70	49.53	38.83	/
(3) Hybrid Family (Ours)				
HySeg_R18	80.83	29.12	15.32+13.57	54.48
HySeg_R50	76.25	57.04	15.32+29.27	54.74
HySeg_LSK_S	61.06	36.88	15.32+15.97	55.37
HySeg_ConvNeXt_B	50.63	183.36	15.32+90.62	55.63

largest gain over fixed values (0.5–2.0), demonstrating the advantage of adaptive neighborhood sensitivity. Within the P2A module, $K = 7$ achieves the best balance between accuracy and efficiency among tested sizes (3, 5, 7, 9). Varying the number of attention heads (2, 4, 8, 16) shows that eight heads offer the most favorable trade-off between performance and computation. These results emphasize the importance of tuning σ, K, and attention heads to fully exploit HySeg's segmentation accuracy and efficiency.

Ablation of Performance–Efficiency Trade-off. As summarized in Tab. 6, we compare discriminative, generative, and hybrid paradigms in terms of FPS, GFLOPs, parameters, and mIoU. Purely discriminative models deliver the highest throughput but at the cost of accuracy: UNetFormer–ResNet18 reaches 188.12 FPS yet only 52.42 mIoU, while LSKNet_S provides a slightly better trade-off (54.01 mIoU with 14.39 M parameters). In contrast, conditional diffusion/flow models are used only as *prior generators*; even with 10-NFE sampling they require up to 527.14 GFLOPs and 320.17 M parameters and run at 5.03–10.24 FPS, underscoring the efficiency gap. Even with DPM-Solver acceleration, UNet2DModel + DPM-Solver (1-NFE) reaches only 40.70 FPS with 49.53 GFLOPs. Our hybrid HySeg family reconciles this gap: for instance, HySeg–ResNet18 attains 54.48 mIoU at 80.83 FPS, combining generative-level accuracy with discriminative-level speed. Each variant integrates a lightweight MeanFlow predictor (15.32 M parameters, 1 NFE) for fast structural inference, introducing negligible latency in the second stage. Beyond numeric gains, this design demonstrates that injecting generative priors need not compromise real-time performance, pointing toward a scalable path for structurally aware segmentation in resource-constrained settings.

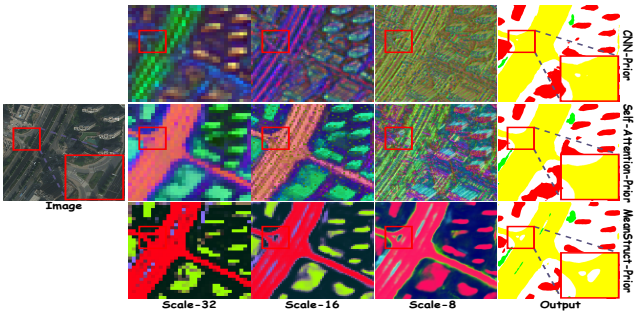


Figure 5. **Visual comparison** of segmentation results using different priors: CNN prior, self-attention prior, and our MeanStruct prior. Highlighted regions show that the MeanStruct prior yields sharper boundaries and stronger topological coherence, especially in complex structures such as the overpass (red box), demonstrating its effectiveness in resolving semantic ambiguities and preserving structural relationships.

Qualitative Representation Analysis. Feature representations learned under three priors, *i.e.,* CNN, self-attention, and MeanStruct, are compared across spatial scales of 32, 16, and 8, as illustrated in Fig. 5. Each prior exhibits distinct structural behavior. The CNN prior emphasizes fine-grained textures but suffers from limited receptive fields, resulting in fragmented, texture-biased activations. The self-attention prior broadens contextual awareness and models long-range dependencies, yet its multi-scale fusion remains coarse with blurred boundaries. In contrast, the proposed MeanStruct prior constructs a hierarchical pyramid that jointly encodes global topology and inter-scale coherence, yielding sharper boundaries and more semantically consistent features. This advantage is especially evident in complex regions (e.g., the overpass in the red box), where MeanStruct preserves connectivity and accurately delineates overlapping components. By embedding topology-aware relationships across scales, it produces continuous and structurally faithful representations that outperform other priors on complex real-world layouts.

5. Conclusions

We introduced HySeg, a hybrid generative–discriminative framework that reformulates remote sensing segmentation as posterior inference guided by generative structural priors. By coupling the MeanStruct prior learner with the P2A and DAS modules, HySeg bridges appearance perception and structural reasoning. This design enables topology-aware priors learned via generative transport to guide discriminative inference, yielding spatially coherent and semantically consistent predictions. Extensive experiments on four benchmarks demonstrate its strong accuracy, robustness, and efficiency. Overall, HySeg represents a promising step toward unifying generative structure learning with discriminative visual understanding.

Acknowledgment. This work was supported in part by the Natural Science Foundation of Fujian Province under Grants 2020J01571 and 2016J01281; in part by the Science and Technology Innovation Special Fund of Fujian Agriculture and Forestry University under Grant KFB25036A; and in part by the Research Fund of Fujian University of Technology under Grant GY-Z23210.

References

[1] Peri Akiva, Matthew Purri, and Matthew Leotta. Self-supervised material and texture representation learning for remote sensing tasks. In *CVPR*, 2022. 2

[2] Hu Cao, Yueyue Wang, Joy Chen, Dongsheng Jiang, Xiaopeng Zhang, Qi Tian, and Manning Wang. Swin-Unet: UNet-like pure transformer for medical image segmentation. In *ECCV*, 2022. 1, 2

[3] Qinglong Cao, Yuntian Chen, Chao Ma, and Xiaokang Yang. Open-vocabulary remote sensing image semantic segmentation. *arXiv preprint arXiv:2409.07683*, 2024. 1

[4] Liang-Chieh Chen, George Papandreou, Florian Schroff, and Hartwig Adam. Rethinking atrous convolution for semantic image segmentation. *arXiv preprint arXiv:1706.05587*, 2017. 1, 2

[5] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, Kevin Murphy, and Alan L. Yuille. DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(4):834– 848, 2018. 1, 2

[6] Liang-Chieh Chen, Yukuan Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder–decoder with atrous separable convolution for semantic image segmentation. In *ECCV*, 2018. 6

[7] Seun-An Choe, Ah-Hyung Shin, Keon-Hee Park, Jinwoo Choi, and Gyeong-Moon Park. Open-set domain adaptation for semantic segmentation. In *CVPR*, 2024. 2

[8] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding. In *CVPR*, 2016. 5

[9] Ilke Demir, Krzysztof Koperski, David Lindenbaum, Guan Pang, Jing Huang, Saikat Basu, Forest Hughes, Devis Tuia, and Ramesh Raskar. DeepGlobe 2018: A challenge to parse the earth through satellite images. In *CVPRW*, 2018. 5

[10] Ruiqi Feng, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. On the guidance of flow matching. In *ICML*, 2025. 3

[11] Jun Fu, Jing Liu, Haijie Tian, Yong Li, Yongjun Bao, Zhiwei Fang, and Hanqing Lu. Dual attention network for scene segmentation. In *CVPR*, 2019. 6

[12] Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. *arXiv preprint arXiv:2505.13447*, 2025. 2, 3, 4

[13] David Griffiths and Jan Boehm. Improving public data for building segmentation from convolutional neural networks CNNs for fused airborne lidar and image data using active contours. *ISPRS Journal of Photogrammetry and Remote Sensing*, 154:70–83, 2019. 1

[14] Yi Guo, Wei Wang, Zhihang Yuan, Rong Cao, Kuan Chen, Zhengyang Chen, Yuanyuan Huo, Yang Zhang, Yuping Wang, Shouda Liu, and Yuxuan Wang. SplitMeanFlow: Interval splitting consistency in few-step generative modeling. *arXiv preprint arXiv:2507.16884*, 2025. 2

[15] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 5

[16] Huiyang Hu, Peijin Wang, Hanbo Bi, Boyuan Tong, Zhaozhi Wang, Wenhui Diao, Hao Chang, Yingchao Feng, Ziqi Zhang, Yaowei Wang, Qixiang Ye, Kun Fu, and Xian Sun. RS-vHeat: Heat conduction guided efficient remote sensing foundation model. In *ICCV*, 2025. 2

[17] Xiaoling Hu, Fuxin Li, Dimitris Samaras, and Chao Chen. Topology-preserving deep image segmentation. In *NeurIPS*, 2019. 5

[18] Shiqi Huang, Shuting He, Huaiyuan Qin, and Bihan Wen. SCORE: Scene context matters in open-vocabulary remote sensing instance segmentation. In *ICCV*, 2025. 2

[19] Chanyoung Kim, Woojung Han, Dayun Ju, and Seong Jae Hwang. EAGLE: Eigen aggregation learning for object-centric unsupervised semantic segmentation. In *CVPR*, 2024. 2

[20] Marcel Kollovieh, Marten Lienen, David Lüdke, Leo Schwinn, and Stephan Günnemann. Flow matching with gaussian process priors for probabilistic time series forecasting. In *ICLR*, 2025. 3

[21] Kaiyu Li, Ruixun Liu, Xiangyong Cao, Xueru Bai, Feng Zhou, Deyu Meng, and Zhi Wang. SegEarth-OV: Towards training-free open-vocabulary segmentation for remote sensing images. In *CVPR*, 2025. 1, 2

[22] Rui Li, Shunyi Zheng, Ce Zhang, Chenxi Duan, Jianlin Su, Libo Wang, and Peter M. Atkinson. Multiattention network for semantic segmentation of fine-resolution remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 60:5607713, 2021. 6

[23] Xin Li, Fan Yang, Ao Luo, Zhicheng Jiao, Hong Cheng, and Zicheng Liu. EFRNet: Efficient feature reconstructing network for real-time scene parsing. *IEEE Transactions on Multimedia*, 24:2852–2865, 2022. 2

[24] Yuxuan Li, Xiang Li, Yimain Dai, Qibin Hou, Yongxiang Liu Li Liu, Ming-Ming Cheng, and Jian Yang. LSKNet: A foundation lightweight backbone for remote sensing. *IJCV*, 133:1410–1431, 2025. 5, 6

[25] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *ICLR*, 2023. 3, 4

[26] Sihan Liu, Yiwei Ma, Xiaoqing Zhang, Haowei Wang, Jiayi Ji, Xiaoshuai Sun, and Rongrong Ji. Rotated multi-scale interaction network for referring remote sensing image segmentation. In *CVPR*, 2024. 1, 2

[27] Wang Liu, Puhong Duan, Zhuojun Xie, Xudong Kang, and Shutao Li. Learn from segment anything model: Local region homogenizing for cross-domain remote sensing image segmentation. In *IGARSS*, 2024. 2

[28] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. *arXiv preprint arXiv:2209.03003*, 2022. 3

[29] Yinhe Liu, Sunan Shi, Junjue Wang, and Yanfei Zhong. Seeing beyond the patch: Scale-adaptive semantic segmentation of high-resolution remote sensing imagery based on reinforcement learning. In *ICCV*, 2023. 1, 2

[30] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, 2021. 5

[31] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *CVPR*, 2022. 5

[32] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *CVPR*, 2015. 5

[33] Ao Luo, Fan Yang, Xin Li, Rui Huang, and Hong Cheng. EKENet: Efficient knowledge enhanced network for real-time scene parsing. *Pattern Recognition*, 111:107671, 2021. 2

[34] Ao Luo, Fan Yang, Xin Li, Yuezun Li, Zhicheng Jiao, Hong Cheng, and Siwei Lyu. Robust scene parsing by mining supportive knowledge from dataset. *IEEE Transactions on Neural Networks and Learning Systems*, 34(5):2633–2646, 2023. 2

[35] Ao Luo, Xin Li, Fan Yang, Jiangyu Liu, Haoqiang Fan, and Shuaicheng Liu. FlowDiffuser: Advancing optical flow estimation with diffusion models. In *CVPR*, 2024. 3

[36] Ao Luo, Xin Li, Fan Yang, Yuezun Li, Zhaoquan Yuan, Shan Zhao, Bing Su, and Xiao Wu. Optical flow matching: Reframing optical flow as continuous transport dynamics. In *CVPR*, 2026. 3

[37] Zifei Luo, Wenzhu Yang, Yunfeng Yuan, Ruru Gou, and Xiaonan Li. Semantic segmentation of agricultural images: A survey. *Information Processing in Agriculture*, 11(2):172–186, 2024. 1

[38] Ye Lyu, George Vosselman, Gui-Song Xia, Alper Yilmaz, and Miicheal Ying Yang. UAVid: A semantic segmentation dataset for uav imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 165:108–119, 2020. 5

[39] Xianping Ma, Qianqian Wu, Xingyu Zhao, Xiaokang Zhang, Man-On Pun, and Bo Huang. SAM-assisted remote sensing imagery semantic segmentation with object and boundary constraints. *IEEE Transactions on Geoscience and Remote Sensing*, 62:5636916, 2024. 6

[40] Xiaowen Ma, Rongrong Lian, Zhenkai Wu, Hongbo Guo, Fan Yang, Mengting Ma, Sensen Wu, Zhenhong Du, Wei Zhang, and Siyang Song. LOGCAN++: Adaptive local–global class-aware network for semantic segmentation of remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 63:4404216, 2025. 6

[41] Oscar Manas, Alexandre Lacoste, Xavier Giró i Nieto, David Vazquez, and Pau Rodriguez. Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data. In *ICCV*, 2021. 1

[42] Ruben Pascual, Christian Ayala, Ruben Sesma, Aranzazu Jurio, Daniel Paternain, and Mikel Galar. Speeding-up diffusion models for remote sensing semantic segmentation. *International Journal of Applied Earth Observation and Geoinformation*, 142:104636, 2025. 2

[43] Jie Qiu, Dizuo Cao, Linwei Dai, Xin Li, Fan Yang, Dong Yu, Changying Wang, Zongheng Wen, Youqin Chen, and Jianzhang Chen. Beyond euclidean assumptions: Geometry-aware adaptive routing for remote sensing segmentation. In *AAAI*, 2026. 2

[44] Franz Rottensteiner, Günter Sohn, Jürgen Jung, Markus Gerke, Christian Baillard, Sergio Benitez, and Uwe Breitkopf. International Society for Photogrammetry and Remote Sensing, 2D semantic labeling contest. Accessed: Mar. 1, 2024. 5

[45] Junao Shen, Qiyun Hu, Tian Feng, Xinyu Wang, Hui Cui, Sensen Wu, and Wei Zhang. In2nect: Inter-class and intra-class neural collapse tuning for semantic segmentation of imbalanced remote sensing images. In *AAAI*, 2025. 6

[46] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich Bauer, and Bjoern H. Menze. cldice – a novel topology-preserving loss function for tubular structure segmentation. In *CVPR*, pages 16560–16569, 2021. 5

[47] Pengfei Song, Jinjiang Li, Zhiyong An, Hui Fan, and Limwei Fan. CTMFNet: Cnn and transformer multiscale fusion network of remote sensing urban scene imagery. *IEEE Transactions on Geoscience and Remote Sensing*, 61:5900314, 2022. 6

[48] Sagar Soni, Akshay Dudhane, Hiyam Debary, Mustansar Fiaz, Muhammad Akhtar Munir, Muhammad Sohail Danish, Paolo Fraccaro, Campbell D Watson, Levente J Klein, Fahad Shahbaz Khan, and Salman Khan. EarthDial: Turning multi-sensory earth observations to interactive dialogues. In *CVPR*, 2025. 2

[49] Vlatko Spasev, Ivica Dimitrovski, Ivan Kitanovski, and Ivan Chorbev. Semantic segmentation of remote sensing images: Definition, methods, datasets and applications. In *International Conference on ICT Innovations*, 2024. 1

[50] Romain Strudel, Ruben Garcia, Ivan Laptev, and Cordelia Schmid. Segmenter: Transformer for semantic segmentation. In *ICCV*, 2021. 6

[51] Nico Stucki, Johannes C. Paetzold, Suprosanna Shit, Bjoern Menze, and Ulrich Bauer. Topologically faithful image segmentation via induced matching of persistence barcodes. In *International Conference on Machine Learning*, pages 32678–32701, 2023. 5

[52] Sumanth Udupa, Prajwal Gurunath, Aniruddh Sikdar, and Suresh Sundaram. MRFP: Learning generalizable semantic segmentation from sim-2-real with multi-resolution feature perturbation. In *CVPR*, 2024. 2

[53] Fengxiang Wang, Hongzhen Wang, Di Wang, Zonghao Guo, Zhenyu Zhong, Long Lan, Wenjing Yang, and Jing Zhang. Harnessing massive satellite imagery with efficient masked image modeling. In *ICCV*, 2025. 1

[54] Jingdong Wang, Ke Sun, Tianheng Cheng, Borui Jiang, Chaorui Deng, Yang Zhao, Dong Liu, Yadong Mu, Mingkui

Tan, Xinggang Wang, Wenyu Liu, and Bin Xiao. Deep high-resolution representation learning for visual recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(10):3349–3364, 2020. 1, 2

[55] Junjue Wang, Zhuo Zheng, Ailong Ma, Xiaoyan Lu, and Yanfei Zhong. LoveDA: A remote sensing land-cover dataset for domain-adaptive semantic segmentation. In *NeurIPS*, 2021. 5

[56] Libo Wang, Rui Li, Chenxi Duan, Ce Zhang, Xiaoliang Meng, and Shenghui Fang. A novel transformer-based semantic segmentation scheme for fine-resolution remote sensing images. *IEEE Geoscience and Remote Sensing Letters*, 19:6506105, 2022. 5, 6

[57] Libo Wang, Rui Li, Ce Zhang, Shenghui Fang, Chenxi Duan, Xiaoliang Meng, and Peter M. Atkinson. UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 190:196–214, 2022. 5, 6

[58] Yanheng Wang, Lianru Gao, Danfeng Hong, Jianjun Sha, Lian Liu, Bing Zhang, Xianhui Rong, and Yonggang Zhang. Mask DeepLab: End-to-end image segmentation for change detection in high-resolution remote sensing images. *International Journal of Applied Earth Observation and Geoinformation*, 104:102582. 1, 2

[59] Bo Wen, Haochen Zhang, Dirk-Uwe G. Bartsch, William R. Freeman, Truong Q. Nguyen, and Cheolhong An. Topology-preserving image segmentation with spatial-aware persistent feature matching. In *CVPRW*, 2025. 6

[60] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. SegFormer: Simple and efficient design for semantic segmentation with transformers. In *NeurIPS*, 2021. 1, 2

[61] Rongtao Xu, Changwei Wang, Jiguang Zhang, Shibiao Xu, Weiliang Meng, and Xiaopeng Zhang. RSSFormer: Foreground saliency enhancement for remote sensing land-cover segmentation. *IEEE Transactions on Image Processing*, 32: 1052–1064, 2023. 2

[62] Yi Zhang, Meng-Hao Guo, Miao Wang, and Shi-Min Hu. Exploring regional clues in CLIP for zero-shot semantic segmentation. In *CVPR*, 2024. 2

[63] Hengshuang Zhao, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia. Pyramid scene parsing network. In *CVPR*, 2017. 1, 2

[64] Jianwei Zhao, Fan Yang, Xin Li, Qiang Zhai, Ao luo, Ziqi Ren, Zhicheng Jiao, and Hong Cheng. Genevar: Causal meanflow for autoregressive gene-to-wsi tile synthesis. In *CVPR*, 2026. 3

[65] Yu Zhong, Xiao Wu, Zihan Cao, Hong-Xia Dou, and Liang-Jian Deng. SSDiff: Spatial–spectral integrated diffusion model for remote sensing pansharpening. In *NeurIPS*, 2024. 2

[66] Hao Zhu, Yan Zhu, Jiayu Xiao, Tianxiang Xiao, Yike Ma, Yucheng Zhang, and Feng Dai. Exact: Exploring space–time perceptive clues for weakly supervised satellite image time series semantic segmentation. In *CVPR*, 2025. 2

[67] Xiaoqian Zhu, Xiangrong Zhang, Tianyang Zhang, Chaowei Fang, Xu Tang, and Licheng Jiao. RegionMatch: Pixel–region collaboration for semi-supervised semantic segmentation in remote sensing. In *IJCAI*, 2025. 1

[68] Xuechao Zou, Kai Li, Junliang Xing, Yu Zhang, Shiying Wang, Lei Jin, and Pin Tao. DiffCR: A fast conditional diffusion framework for cloud removal from optical satellite images. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–14, 2024. 2

[69] Xuechao Zou, Yue Li, Shun Zhang, Kai Li, Shiying Wang, Pin Tao, Junliang Xing, and Congyan Lang. Dynamic dictionary learning for remote sensing image segmentation. In *ICCV*, 2025. 5, 6